

HOW DATA SCIENCE REVOLUTIONIZES THE FUTURE OF HEALTHCARE

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Storytelling with Data | Southern Methodist University

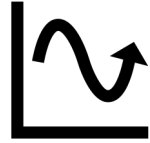


COMMON HEALTHCARE CHALLENGES



COMPLICATED DIGITAL INTEGRATION

Many hospitals still use outdated systems that can't easily share imaging data, limiting remote consultations, second opinions, and team workflows.



LARGE AND COMPLEX MACHINERY

Traditional ultrasound machines are large, room-bound, and require multiple probes - impractical for emergency or mobile use.



HIGH COST

Imaging devices can cost tens of thousands of dollars, making them inaccessible for small clinics, EMS, and global health settings.



LIMITED ACCESS

Rural and underresourced areas lack access to imaging tools, causing delays in diagnosis and poorer health outcomes.



INEFFICIENT EMERGENCY RESPONSE AND PATIENT MOBILITY

Transporting critically ill or trauma patients for imaging is dangerous and time-consuming. Clinicians need tools that come to the patient.

MEET BUTTERFLY

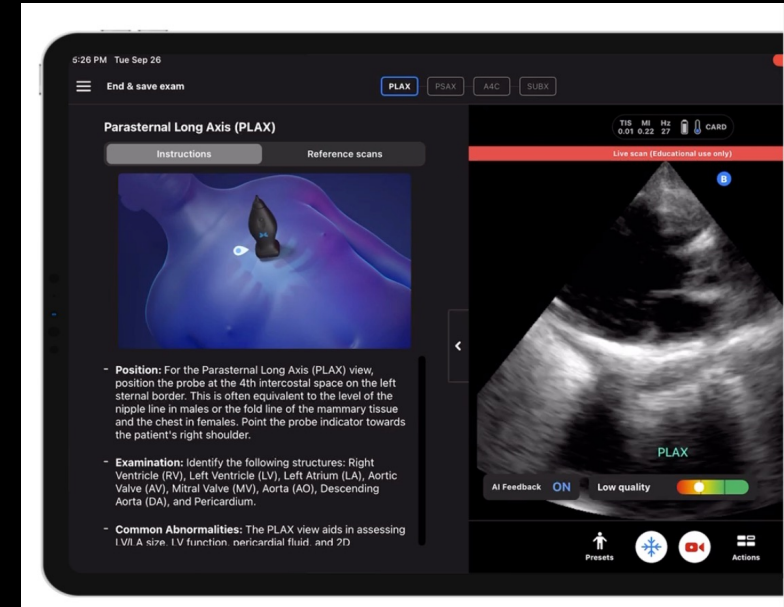
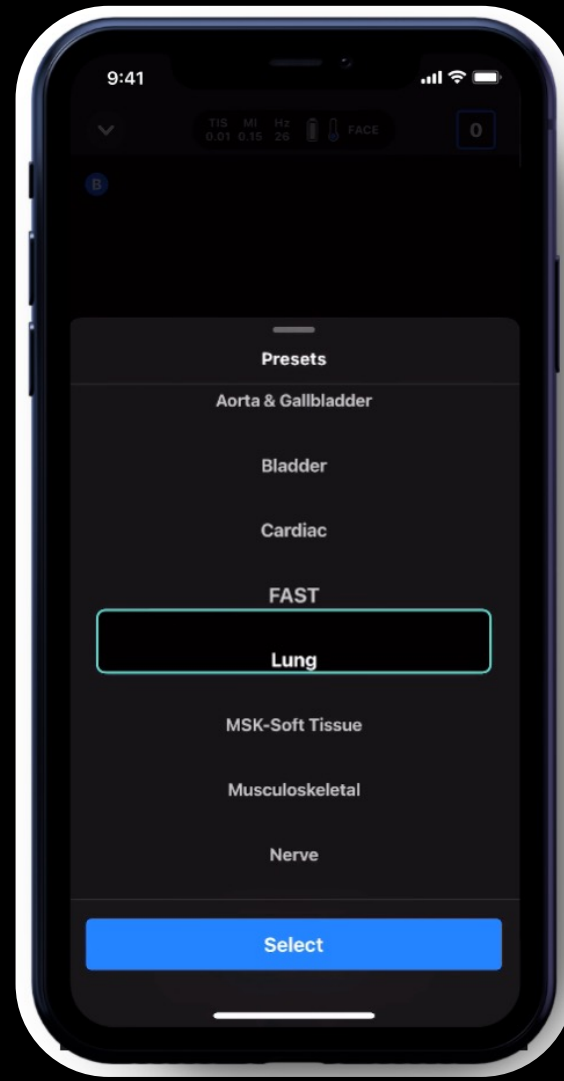
IQ/IQ+/IQ³

- Butterfly iQ is a portable, point-of-care ultrasound (POCUS) device that connects to smartphones or tablets.
- Replaces multiple traditional transducers with a single silicon chip-based probe.
- FDA-cleared for whole-body imaging in a wide range of medical settings (i.e. emergency rooms, rural clinics, etc.).
- Enables AI-powered imaging and real-time diagnostics at the bedside or in the field.



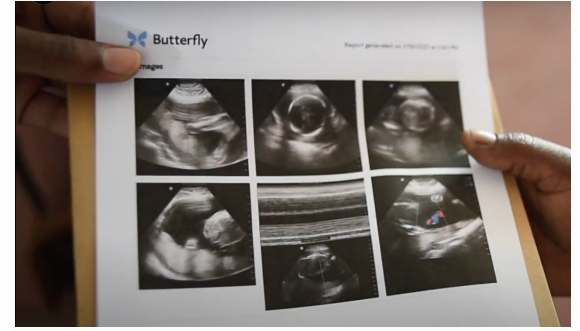
AI BEHIND THE BUTTERFLY IQ+

- AI enhances image acquisition and interpretation, & diagnostic speed
- Trained on 1B+ frames, 3M+ labeled scans
- Real-time guidance for probe positioning and capturing images.
- Automated measurements (e.g., bladder volume) with high precision.
- ScanLab™ simplifies training for new users & reduces the learning curve for non-specialists.



INSIDE THE TECHNOLOGY:

DATA SCIENCE'S ROLE IN BUTTERFLY IQ+



Signal Acquisition

- CMUT transducer captures raw ultrasound signals

Image Reconstruction

- Real-time image formation

User Interface

- Real-time display
- User input and annotations

Data Analytics & Feedback

- Usage analytics
- Model improvement via feedback loops
- Population health insights

Digital Signal Processing

- Algorithms convert signals to digital data
- Noise reduction, artifact removal

AI/ML-Powered Interpretation

- Automated structure recognition
- Automated measurements
- Quality assessment

Cloud Integration

- Secure upload of images and metadata
- Remote access and telemedicine

Security & Compliance

- Data encryption
- HIPAA compliance

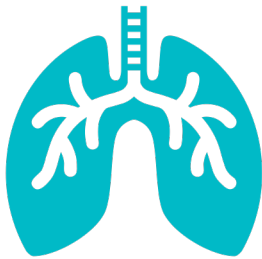
REAL-WORLD CASE:

HENNEPIN EMS CASE STUDY

- 40 ambulances equipped with Butterfly iQ+.
- 210 paramedics trained in its use.
- These paramedics completed over 6,000 scans in 2024.
- 10% of scans significantly changed patient care decisions.
- Enabled early diagnosis of serious conditions like pulmonary embolism and aortic aneurysm.



CLINICAL USE EXAMPLES



Lung scans help detect fluid buildup or pneumonia quickly.



Cardiac imaging identifies issues like heart failure before hospital arrival.

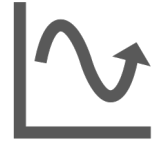


Aortic scans detect life-threatening aneurysms in emergency settings.

BUTTERFLY IQ BRIDGES THE GAP



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HIGH COST



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RESPONSE AND PATIENT
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SELF-SERVICE AND SEAMLESS IMAGING

Secure cloud storage + telemedicine-ready platform enables fast image sharing and remote specialist collaboration.

PORTABLE, ALL-IN-ONE DESIGN

Replaces multiple traditional probes with a single silicon chip; fits in your pocket and connects to a mobile device with multiple preset options.

AFFORDABLE DIAGNOSTIC IMAGING

Lower up-front and maintenance costs thanks to chip-based design, making ultrasound accessible to more providers.
Cost: \$500

EXPANDING IMAGING ACCESS

Portable and battery-powered ideal for fieldwork, rural settings, and global health missions.

ON-THE-SPOT EMERGENCY IMAGING

Allows paramedics and bedside providers to perform real-time scans without moving the patient - critical in trauma situations.

THE FUTURE IS HERE — AND IT'S SMART

THE PERFECT PAIRING OF MAN AND MACHINE TO MAKE AN IMPACT THAT MATTERS



DATA DRIVEN DECISION MAKING

Empowers clinicians to make **faster**, more **accurate diagnoses** at the point of care



ACCESSIBLE AND SCALABLE

Allows for **portability** and **ease of use** to deliver high-quality ultrasound imaging anywhere



RAPID RESPONSE

Enables **quick responses in critical situations**, improving patient outcomes through timely interventions



THANK YOU!
QUESTIONS?

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